

Phase 5 - Hawaii Micro-Study & Empirical Validation

Phase 5 is the empirical anchor of the thesis. The earlier phases produce a national-scale opportunity cost figure derived from federal land-valuation datasets, but the figure has no parcel-level reality check until Phase 5 runs. The Hawaii micro-study integrates the national pipeline outputs with parcel-level cadastral, tax-roll, and zoning data published by the City and County of Honolulu, then asks two questions: how closely does the model's HBU estimate align with what municipal assessors actually report on the tax rolls, and how much of the modeled opportunity cost is bounded by current zoning law?

The phase is organized as two complementary tracks. **Phase 5a** is a manual pilot validation: six high-profile Hawaii golf courses spanning all four counties (Honolulu, Maui, Hawaii, Kauai) are individually compared against their 2022 tax assessment values, which I computed by hand to allow careful per-course interpretation. **Phase 5b** is a six-step automated pipeline implemented in all three languages: it processes every golf course on Oahu at parcel resolution, intersecting OSM golf polygons with the Honolulu cadastre to identify the Tax Map Keys (TMKs) that constitute each course, pooling economic estimates across $\$M = 100\$$ MICE imputations per language via Rubin's Rules, and intersecting each parcel with the Honolulu zoning layer to determine its dominant land-use classification.

The data integration is non-trivial. Honolulu's cadastral and zoning GeoPackages are distributed in EPSG:3760 (NAD83(HARN) Hawaii Zone 3, US survey feet), which must be reprojected to EPSG:5070 to align with the national pipeline. The cadastre contains parent parcels and CPR (Condominium Property Regime) sub-parcels; I designed the pipeline to filter to parent parcels only, since CPRs are unit-level subdivisions of a single physical parcel and would inflate the count. The OSM-derived golf polygons cover only 31.4% of the Phase 1 course points via direct point-in-polygon match, with the remainder requiring spatial intersection against the cadastre to recover the parcels each course occupies.

Key result

The Phase 5a pilot produces an average model-to-assessed ratio of **1.33×** across the six courses, indicating that the national HBU model systematically exceeds going-concern tax assessments by roughly one-third. The ratio is narrowest in urban Honolulu (Waialae Country Club at 1.16×, Turtle Bay Resort at 1.23×) and widest in rural Big Island courses (Kohala Country Club at 1.49×, Hualalai Golf Club at 1.69×). This pattern is consistent with the FHFA urban residential proxy more closely tracking municipal assessment practice than the USDA rural agricultural proxy does, and provides a defensible empirical anchor for the gap between gross HBU and going-concern current-use value.

Phase 5b produces an aggregate Oahu opportunity cost of **\$25.4B** under Rubin's Rules pooling (95% CI: \$22.7B–\$28.1B), distributed across **1,072 unique TMK parcels** comprising 33 deduplicated golf courses. The geographic distribution is striking: **63.2% of the parcels (677 of 1,072) sit in the Ewa District (Zone 9)**, the suburban housing development corridor running through Kapolei, Pearl City, and Ewa Beach. This concentration directly co-locates Oahu's golf footprint with the island's primary vector for new residential development, sharpening the policy stakes of the unrealized opportunity cost.

The most consequential finding emerges from the Step 6 zoning intersection. **81.7% of all Oahu golf land sits within Preservation (P-1, P-2) or Federal/Military (F-1) zones** - areas where residential redevelopment is, under current zoning, statutorily prohibited. Only roughly 4.5% of the footprint sits in Resort, Residential, or Country zones where redevelopment to housing would be legally compatible with existing zoning. A

complementary penetration finding shows that golf occupies **25.4% of all Resort-zoned land on Oahu** - the highest penetration rate of any zone class on the island - indicating that golf is not an incidental amenity within resort zoning but is the dominant land use in that classification.

Together these findings produce what I call the **Preservation Paradox**: the unrestricted gross HBU framework computes large opportunity costs precisely on land that current zoning cannot release, while the small share of golf land in zones permitting redevelopment carries a correspondingly small share of the realizable economic potential.

What was solved

Two integration challenges were load-bearing for the result. First, the OSM golf polygons and the Honolulu cadastre use different coordinate reference systems, different polygon edge conventions, and different feature granularities - OSM treats a golf course as a single multipolygon while the cadastre breaks the same physical course into dozens of TMK parcels. My fix is the cookie-cutter intersection in Step 2, which assigns each parcel its OSM golf overlap acreage and produces a clean parcel-to-course mapping. Second, R's

`sf::st_intersection` reports the P-1 (Restricted Preservation) overlap acreage as 523.5 acres, while Python and Julia report 744.6 acres. The discrepancy reflects edge-case polygon-boundary handling in `sf` versus GDAL/Shapely; I use the Python/Julia results as canonical for the headline figures while retaining R as a cross-check. All other zone classes agree across the three languages to within 0.1 acres.

Outputs flow forward into Phase 6, which generates the Hawaii Gap dumbbell chart, the TMK concentration map, the zoning composition waffle chart, and the other Hawaii-specific visualizations that anchor the thesis's Section 5.4. The Phase 5 findings are also the empirical foundation for the legally-permissible HBU framework introduced in Section 3.3 of the thesis: $V_{\text{OC}}^{\text{legal}} \leq V_{\text{OC}}^{\text{gross}}$, with the gap between the two bounded by the share of acreage in zones that legally permit the unrestricted use.